

FYS5429/9429 April 30

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FYS55429/9429 APRIL 30

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From VAEs to Diffusion models

$$X = \{x_0, x_1, \dots, x_{n-1}\}$$

$$x_i \sim \text{i.i.d.}$$

$$P(X) = \prod_{x_i \in X} P(x_i)$$

Latent space/hidden space

$$h_i \in H$$

$$H = \{h_0, h_1, \dots, h_{M-1}\}$$

$$P(x_i, h_j; \theta) = \frac{f(x_i, h_j; \theta)}{z(\theta)}$$

$$z(\theta) = \sum_{x_i \in X} \sum_{h_j \in H} \frac{f(x_i, h_j; \theta)}{z(\theta)}$$

Marginal distribution

$$p(x_i; \theta) = \sum_{h_j \in H} \frac{f(x_i, h_j; \theta)}{Z(\theta)}$$

$$p(h_j; \theta) = \sum_{x_i \in X} \frac{f(x_i, h_j; \theta)}{Z(\theta)}$$

$$\hat{\theta} = \arg \max_{\theta} \log p(x; \theta)$$

$$\nabla_{\theta} \log p(x; \theta)$$

Restricted BM

$$\nabla_{\theta} \log p(x; \theta) =$$

$$\nabla_{\theta} \left[ \sum_{x_i' \in X} \log f(x_i'; \theta) \right]_{\text{Data}}$$

$$- \mathbb{E}_{x_i' \sim X} \left[ \nabla_{\theta} \log f(x_i'; \theta) \right]_{\text{Model}}$$

$$x_i' \sim p(x_i'; \theta) \quad \text{Reconstruction}$$

Equivalent with Kullback-Leibler minimization (KL)

$$D_{KL}(P_{data} \parallel P_E) =$$

$$\sum_x P_{data}(x) \log \frac{P_{data}(x)}{P_E(x)}$$

$$= \sum_x P_{data}(x) \log P_{data}(x) - \sum_x P_{data}(x) \log P_E(x)$$

$$D_{KL}(P||Q) \geq 0$$

equality if  $P = Q$

$D_{KL}$  is not symmetric

$$D_{KL}(P||Q) \neq D_{KL}(Q||P)$$

$D_{KL}(P||Q)$  penalizes regions  
where  $p > 0$  but  $q \approx 0$

VAES

$$x_i' \rightarrow x \quad \wedge \quad h_j \rightarrow h$$

$$P(x; \theta) = P_{\theta}(x)$$

$$P_{\theta}(x) = P_{\theta}(x, h) / P_{\theta}(h|x)$$

$$\boxed{P_{\theta}(x)} = \int_{h \in H} dh \, P_{\theta}(x, h)$$

$$= \int dh \, P_{\theta}(x|h) P_{\theta}(h)$$



Latent space

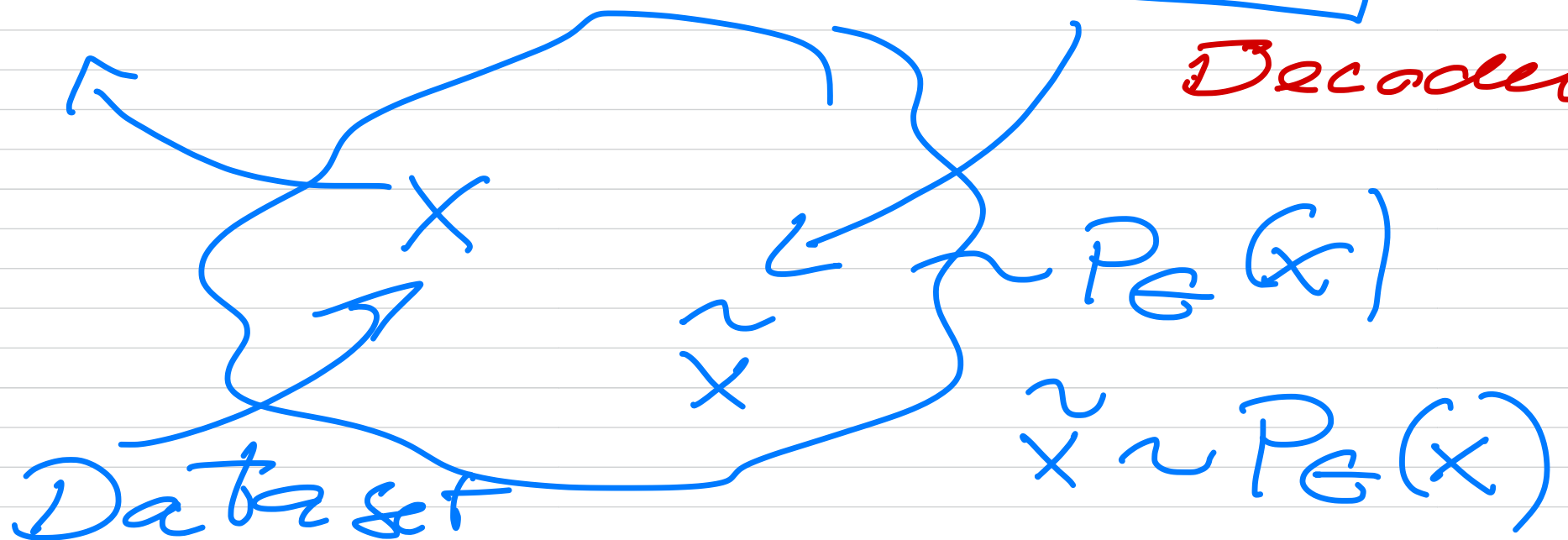
$$P_{\theta}(h)$$

$$q_{\phi}(h|x)$$

Encoder

$$P_{\theta}(x|h)$$

Decoder



We want  $\log$  of  $P_\theta(x)$   
and its gradient

$$\int q_\phi(h|x) dh = 1$$

$$\log P_\theta(x) = \log P_\theta(x) \int q_\phi(x|h) \\ \times dh$$

$$= \int q_\phi(h|x) \log P_\theta(x) dh$$

$$= E_{h \sim q_\phi} [\log P_\theta(x)]$$

$$= E_{h \sim q_\phi} \left[ \log \frac{p_\theta(x, h)}{p_\theta(h|x)} \right] \overset{\text{1}}{\uparrow}$$

$$\underline{1} = \frac{q_\phi(h|x)}{q_\phi(h|x)}$$

$$= E_{h \sim q_\phi} \left[ \log \left( \frac{p_\theta(x, h) q_\phi(h|x)}{p_\theta(h|x) q_\phi(h|x)} \right) \right]$$

$$= E_{h \sim q_\phi} \left[ \log \frac{p_\theta(x, h)}{q_\phi(h|x)} \right]$$

$$+ E_{h \sim q_\phi} \left[ \log \frac{q_\phi(h|x)}{p_\theta(h|x)} \right]$$

$$\underbrace{\hspace{15em}}_{D_{KL}(q_\phi(h|x) || p_\theta(h|x))} \\ (\geq 0)$$

$$= E_{h \sim q_\phi} \left[ \log \frac{P_\theta(x|h)}{q_\phi(h|x)} \right]$$

ELBO = evidence lower bound

$$+ D_{KL}(q_\phi(h|x) || P_\theta(h|x))$$

$$P_\theta(x, h) = P_\theta(x|h) P_\theta(h)$$

$$P_\theta(x) = \int dx P_\theta(x|h) P_\theta(h)$$

$$= E_{h \sim q_\phi} \left[ \log \frac{p_\theta(x|h) p_\theta(h)}{q_\phi(h|x)} \right]$$

$$= E_{h \sim q_\phi} \left[ \log p_\theta(x|h) \right]$$

$$+ E_{h \sim q_q} \left[ \log \frac{p_\theta(h)}{q_\phi(h|x)} \right]$$

$$= E_{h \sim q_\phi} \left[ \log p_\theta(x|h) \right] - D_{KL}(q_\phi(h|x) || p_\theta(h))$$

$q_{\phi}(h|x)$  is treated as an encoder

$q_{\phi}(h|x) \sim \mathcal{N}(h, \mu_{\phi}(x); \sigma_{\phi}^2)$   
multivariate Gaussian

$p_{\phi}(h) = p(h) \sim \mathcal{N}(h; 0, 1)$

# Diffusion Models & Markov chains

$$P_i(t) = \sum_j T_{ij} P_j(t')$$

Transition probab

$$P(t) = \begin{bmatrix} P_0(t) \\ P_1(t) \\ \vdots \\ P_{n-1}(t) \end{bmatrix} \quad \sum_i P_i(t) = 1$$



$$\sum_j \overline{T_{ij}} = 1$$

$$|\lambda_{\max}(T)| = 1$$

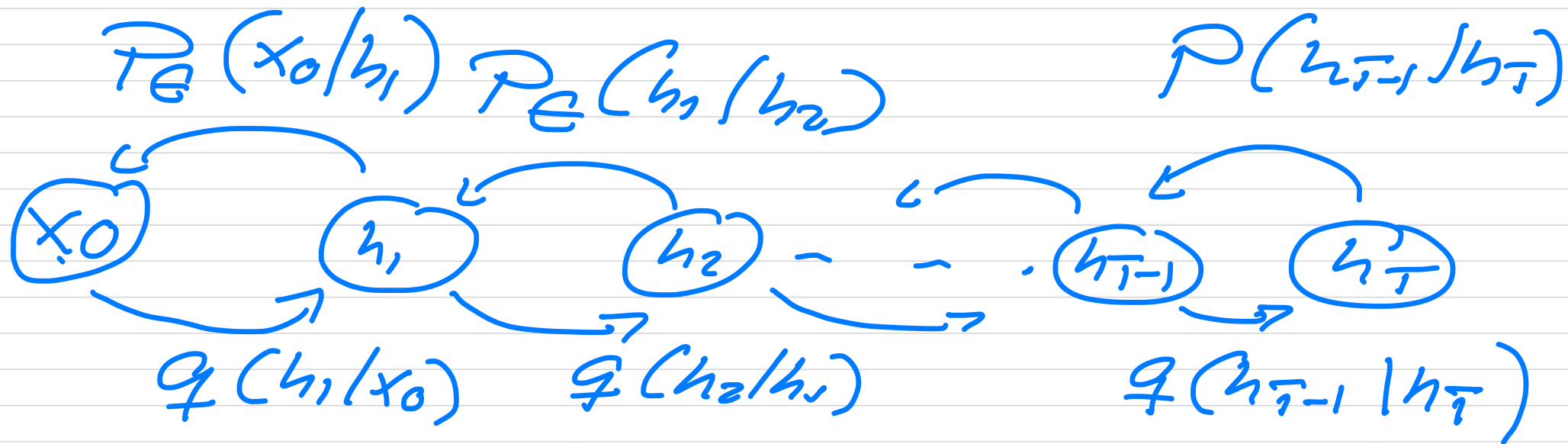
repeat many times

$$\vec{p}(t_n) = T^n \vec{p}(t_0)$$

$\lim_{t_n \rightarrow \infty}$

$$\vec{p} = \frac{1}{1} \vec{p}$$

# Diffusion model



$$\tilde{x}_0 \sim P_\theta(x_0)$$